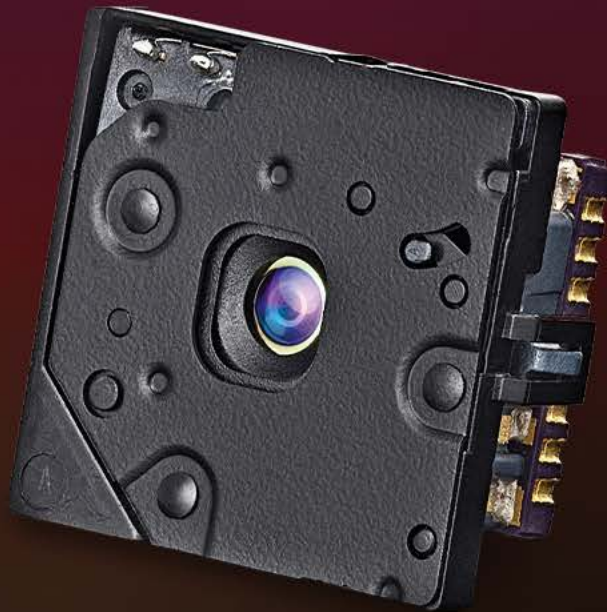
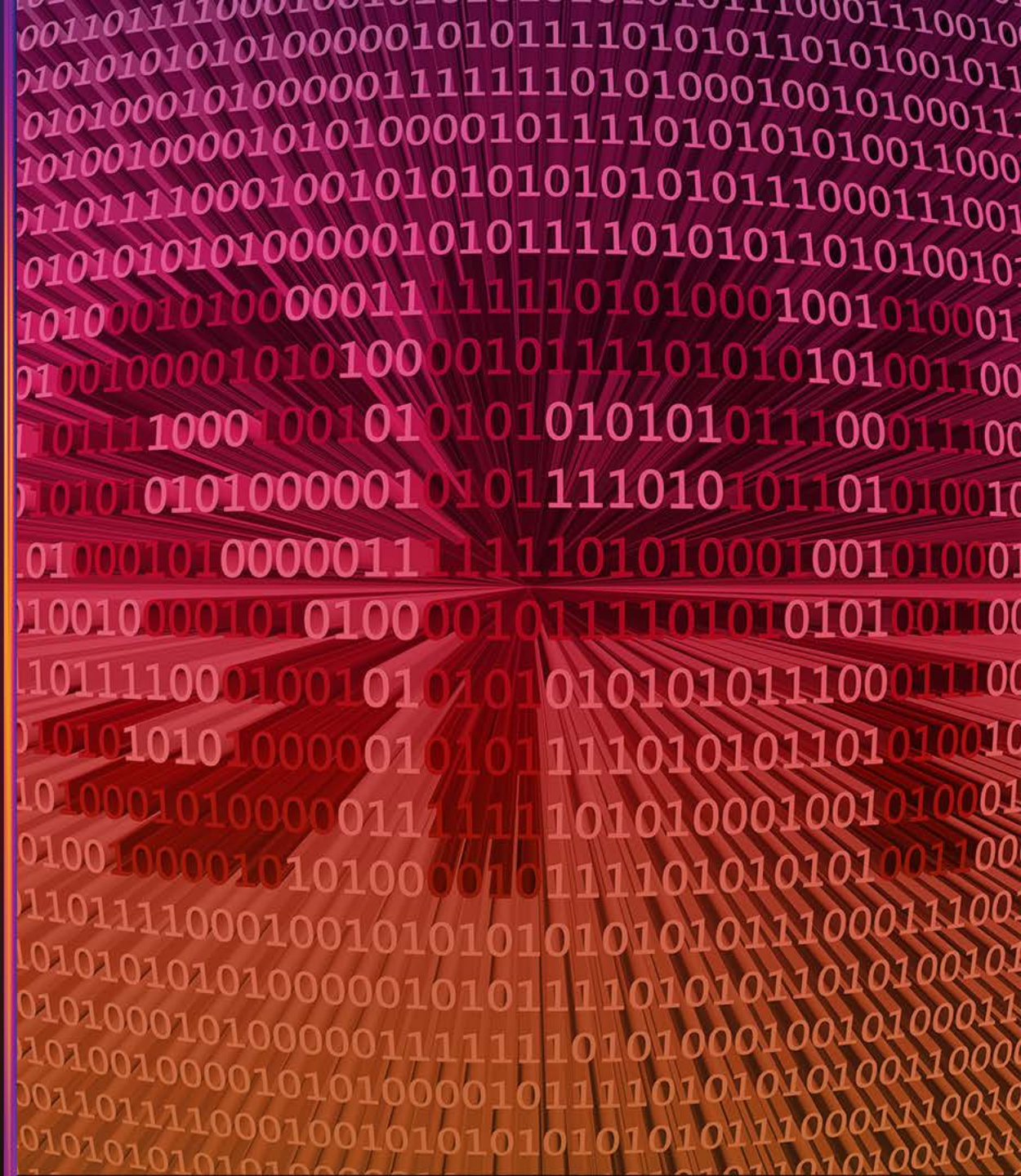


SPI AND I2C WITH FLIR LEPTON[®]

**THERMAL
INTEGRATION
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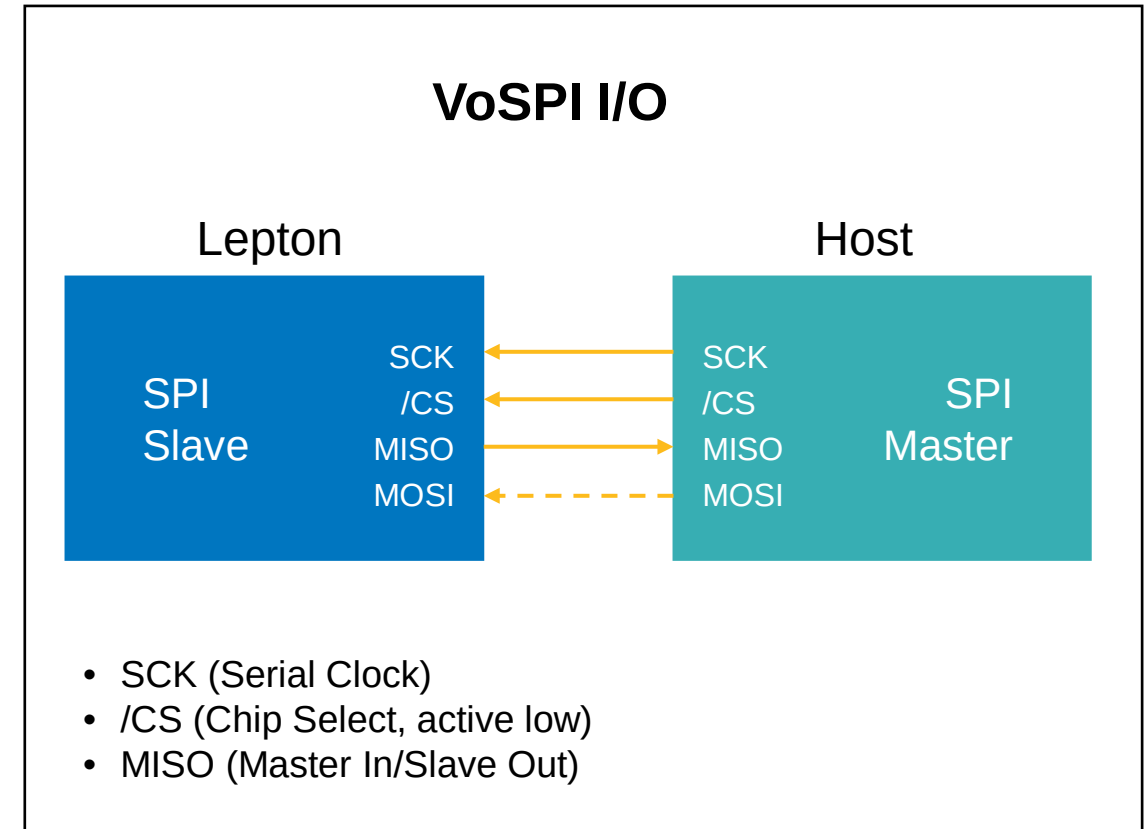


Approved for public release: Teledyne FLIR approved [FLIRGTC-SBA-015]



SPI and I2C Connections

- Level shifting may be necessary
- I2C requires 16-bit registers and transfers
- I2C and SPI use VDDIO voltage levels
- Lepton requires a dedicated SPI bus
- A dedicated I2C bus for Lepton is also recommended



Lepton VoSPI Driver



The Lepton 2.5 sends out frames at a rate of 27Hz, but 2 of 3 frames are duplicates

- All must be captured to maintain synchronization



Lepton 3.5 sends out segments (quarters of full frames) at 110Hz, and 2 of 6 frames are duplicates so the segments must be assembled to create a full image

- Run the master SPI clock at $\leq 24\text{MHz}$ to provide extra room for latency
- Maintain timing to prevent loss of synchronization, which can stop the video stream

Full descriptions of the VoSPI interface, data format, and timing are available in the VoSPI developers guide

Re-establish sync logic

- Assert the CS pin and idle SCK pin for 5 frames (or segments) to reset VoSPI stream

Lepton VoSPI Driver

Reference drivers available on different hardware available on www.flir.com/leptonintegration

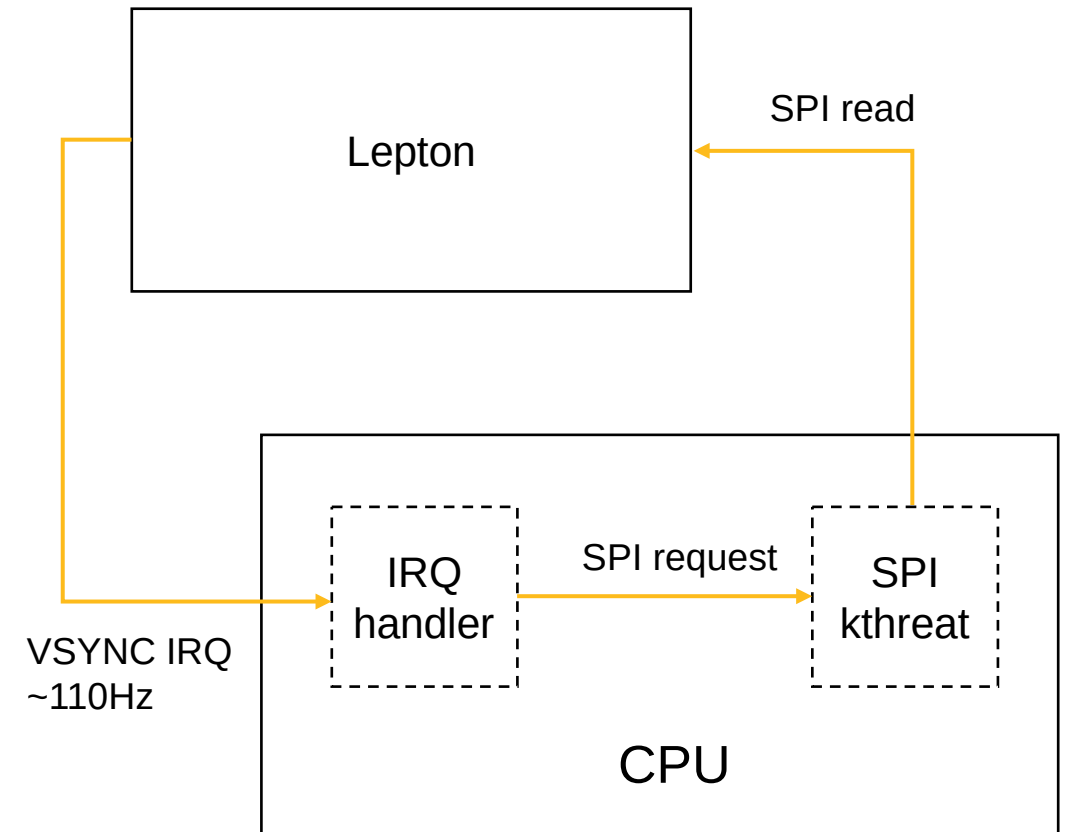
- Raspberry Pi 2-3
- Beaglebone Black
- Amberella Processors
- Qualcomm 600 and 800 series

VSYNC GPIO highly recommended for Lepton 3.5

- Requires DMA to work on most processors (process scheduler must be overridden)
- LEP_SetOemGpioMode(fd, LEP_OEM_GPIO_MODE_VSYNC);
- Interrupt driven VoSPI ensures timing does not get out of sync over time
- Requires connection to VSYNC on Lepton for hardware interrupt

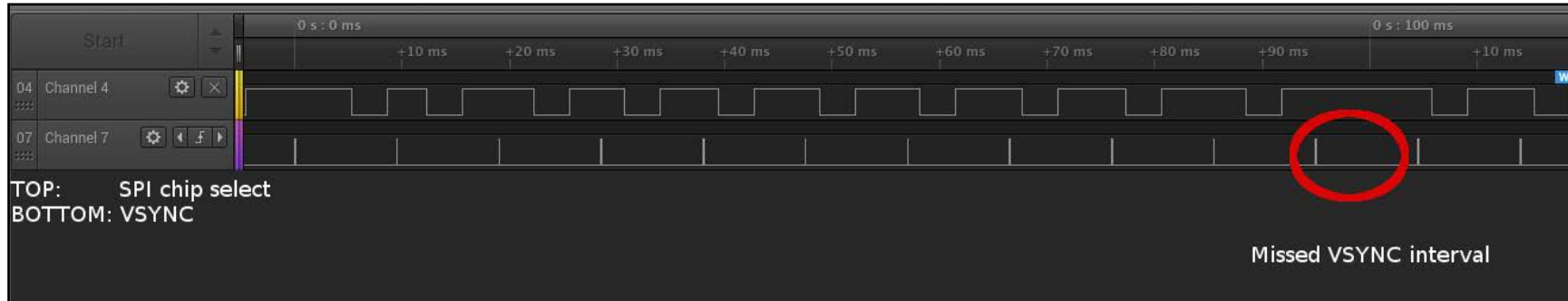
Processor + Lepton 3.5

- Lepton Data rate
 - 164bytes x 60packets @ ~110Hz
 - = ~1MB/sec
- If CPU loses sync, Lepton will output discard (0xFF) (bad) data
- Hardware must be consistent and have low latency and, in some cases, a microcontroller can help if optimization is an issue
- If the process scheduler interrupts SPI read or GPIO Vsync triggers for >3ms, synchronization may be lost

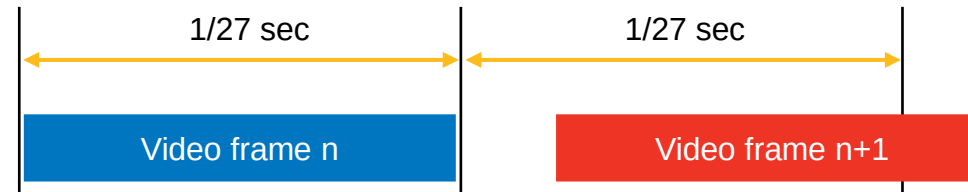


Lepton 3.5 VoSPI Signal Timing

WORST



Intra-Frame Delay Too Long – Failure to Read Out an Entire Frame Before the Next is Available

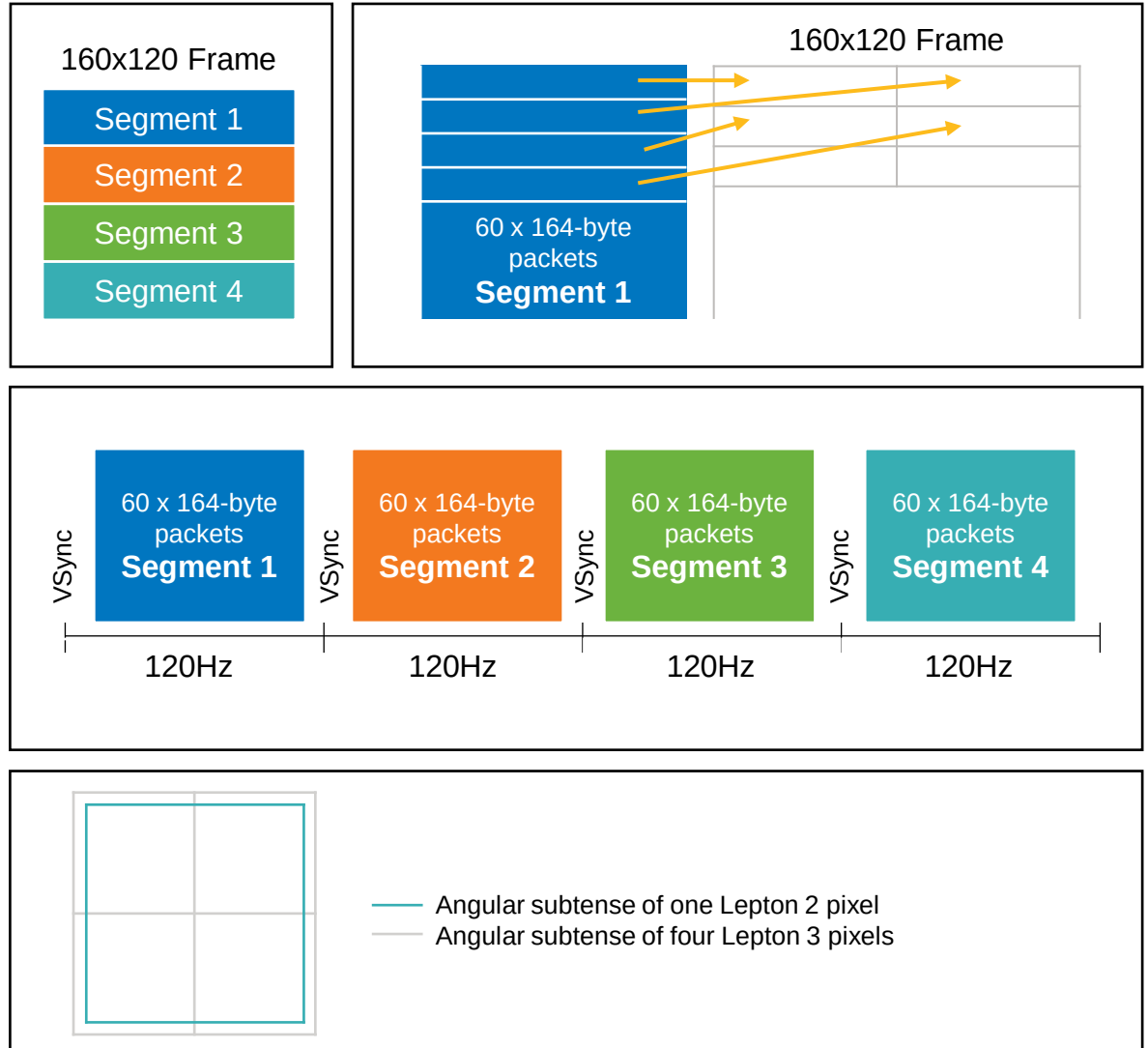


Master CS pin must follow VSYNC GPIO with < 3ms latency

BEST

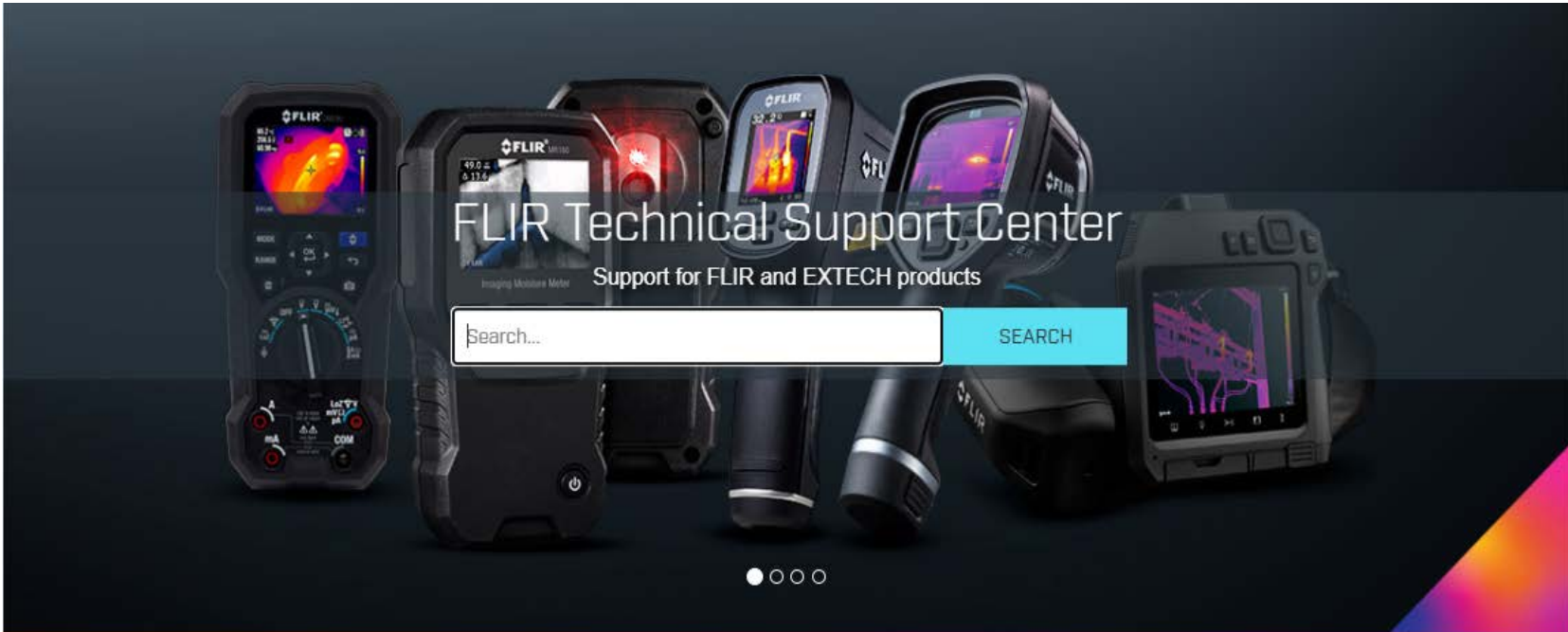
Lepton 2.5 vs. Lepton 3.5

- Lepton 3.5 has 4x the pixels at 160x120 versus 80x60 for Lepton 2.5
- Data Rate is 4X greater for Lepton 3.5
SPI clock rate is 4X faster, making SPI video timing stricter
- Lepton 3.5 transmits video in segments instead of full frames
- Segments require stitching to assemble full frames
- Refer to Lepton 3.5 VoSPI Developer Guide Doc #110-0144-50
- I2C protocol is the same



Support





FLIR Technical Support Center
Support for FLIR and EXTECH products

Search... SEARCH

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Additional integration resources are available at: www.flir.com/leptonintegration


[Ask a Question](#)

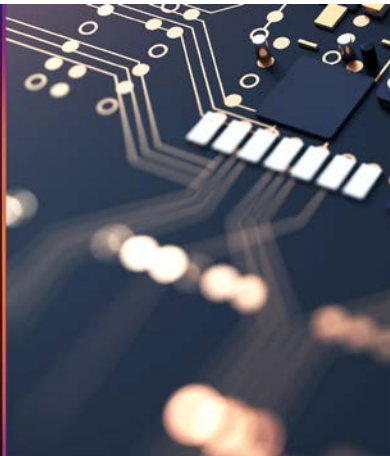
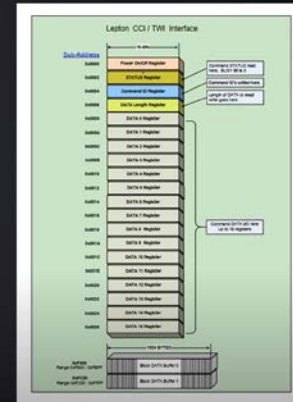
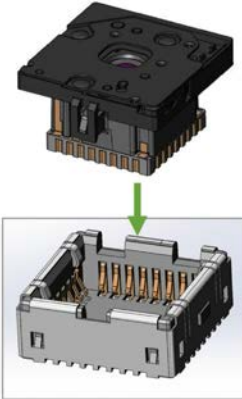

[Software, SDK and Firmware](#)


[Datasheets and FOV Calculators](#)


[User Manuals and Product Drawings](#)

All drawings and CAD are available via www.flir.custhelp.com

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Thank you.

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